Precision medicine and quantitative imaging in glioblastoma

ELMED219 (Artificial Intelligence and Computational Medicine)
BMED365 (Computational imaging, modelling and AI in biomedicine)
January 5-30, 2026

https://github.com/arvidl/ELMED219-2026

https://github.com/arvidl/BMED365-2026

Team #k

Team #k members: (b_1, b_2, b_3, e_1, e_2, e_3)

John Smith b_1)
. . .
Mary Johnson (e_3)

Report Organization

Research Plan

(3 to 5 pages including figures and reference list)

Section	**Content**	**Questions to Consider**
Background	Brief introduction to the field	What is the clinical problem? What are current approaches?
Objectives	Aims and expected outcomes	What do you want to achieve? Why is it relevant?
Materials	Data sources, patient cohorts	Which datasets would you use? What kind of images?
Methods	Imaging analysis, ML approaches	How would you process images? Which methods would you apply?
Evaluation	How to assess results	How would you know if your approach works?

Assessment Criteria

Component	**Weight**	**What We Look For**
Background and Objectives	20%	Clear description of the problem, well-defined aims, understanding of clinical relevance
Materials and Methods	30%	Appropriate choice of data and methods, logical workflow, demonstration of understanding
Relevance and Impact	20%	Why this project matters, potential benefits, realistic scope
Ethics and Data Management	20%	Awareness of privacy and ethical issues, basic data management plan, FAIR principles
Writing Quality	10%	Clear structure, appropriate use of references, readable figures, good language

Note: We do not expect you to propose something completely novel or to have deep technical expertise. We are looking for evidence that you understand the field, can describe a reasonable approach, and have thought carefully about the ethical dimensions.

1 Research plan

1.1 A brief background to the field

Glioblastoma is ... [1]. ... Aldape et al. [2] ... DUMMY TEXT: Fusce mauris. Vestibulum luctus nibh at lectus. Sed bibendum, nulla a faucibus semper, leo velit ultricies tellus, ac venenatis arcu wisi vel nisl. Vestibulum diam. Aliquam pellentesque, augue quis sagittis posuere, turpis lacus congue quam, in hendrerit risus eros eget felis. Maecenas eget erat in sapien mattis porttitor. Vestibulum porttitor. Nulla facilisi. Sed a turpis eu lacus commodo facilisis. Morbi fringilla, wisi in dignissim interdum, justo lectus sagittis dui, et vehicula libero dui cursus dui. Mauris tempor ligula sed lacus. Duis cursus enim ut augue. Cras ac magna. Cras nulla. Nulla egestas. Curabitur a leo. Quisque egestas wisi eget nunc. Nam feugiat lacus vel est. Curabitur consectetuer.

1.2 Objectives and expected impact

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1.3 Material and methods

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Table is given in Tab. 1 ...

header 1	header 2	header 3
cell1	cell2	cell3
cell4	cell5	cell6
cell7	cell8	cell9

Table 1: This is an ELMED219 dummy table.

Data is illustrated in Fig. 1 ...

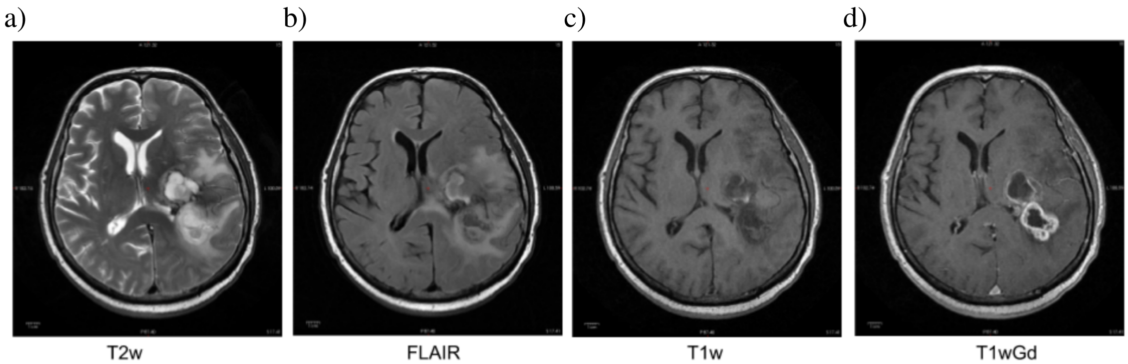

Figure 1: This is an ELMED219 / BMED365 dummy figure. (a) T2w, (b) FLAIR, (c) T1w, (d) T1wGd. (Image taken from study TCGA-06-1802 in the TCGA-GBM data collection [ref [2] in Section 2])

1.4 Evaluation

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References

[1] Louis DN, Perry A, Wesseling P et al. The 2021 WHO Classification of Tumors of the Central Nervous System: a summary. Neuro-Oncology 2021;23(8):1231–1251.
`https://doi.org/10.1093/neuonc/noab106`

[2] Aldape K, Brindle KM, Chesler L et al. Challenges to curing primary brain tumours. Nat Rev Clin Oncol 2019;16:509–520.
`https://doi.org/10.1038/s41571-019-0177-5`

2 Data management plan and ethical considerations

2.1 Description of generated data and code

We will be using data from the TCGA-GBM data collection [2] and The University of California San Francisco Preoperative Diffuse Glioma MRI (UCSF-PDGM) collection [3, 1] provided with a CC BY 4.0 License.

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2.2 Sharing of data and code

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2.3 Ethical considerations

According to Vollmer et al. [4] ... DUMMY TEXT: Suspendisse vitae elit. Aliquam arcu neque, ornare in, ullamcorper quis, commodo eu, libero. Fusce sagittis erat at erat tristique mollis. Maecenas sapien libero, molestie et, lobortis in, sodales eget, dui. Morbi ultrices rutrum lorem. Nam elementum ullamcorper leo. Morbi dui. Aliquam sagittis. Nunc placerat. Pellentesque tristique sodales est. Maecenas imperdiet lacinia velit. Cras non urna. Morbi eros pede, suscipit ac, varius vel, egestas non, eros. Praesent malesuada, diam id pretium elementum, eros sem dictum tortor, vel consectetuer odio sem sed wisi.

References

[1] Calabrese E, Villanueva-Meyer J, Rudie J, Rauschecker A, Baid U, Bakas S, Cha S, Mongan J, Hess, C (2022). The University of California San Francisco Preoperative Diffuse Glioma MRI (UCSF-PDGM) (Version 2) [Dataset]. The Cancer Imaging Archive. DOI: https://doi.org/10.7937/tcia.bdgf-8v37

[2] The TCGA-GBM data collection. Accessed on 01/05/2026.
https://wiki.cancerimagingarchive.net/pages/viewpage.action?pageId=1966258

[3] The University of California San Francisco Preoperative Diffuse Glioma MRI (UCSF-PDGM) data collection. Accessed on 01/05/2026.
https://wiki.cancerimagingarchive.net/pages/viewpage.action?pageId=119705830

[4] Vollmer S, Mateen BA, Bohner G et al. Machine learning and artificial intelligence research for patient benefit: 20 critical questions on transparency, replicability, ethics, and effectiveness. BMJ 2020;368:l6927.
https://www.bmj.com/content/368/bmj.l6927